Soundness of modal tableau derivability

The Open Logic Project contributors

openlogic:normal-modal-logic/tableaux:local-snapshot-20261013;
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Editorial note. The counted statement is the author corollary labelled cor:entailment-soundness (2153-2155); the proof span 2157-2170 is the author proof environment that immediately follows it.

Editorial note. The proof's closing sentence prints 'Gamma proves A' where 'Gamma entails A' is meant; the unit is reproduced without mathematical changes, so the slip is kept as authored.

Editorial note. External reference numbers render the unit-internal labels with hyphens in place of colons (pipeline number charset); the Source reference lines below the body give each label's exact source span.

Editorial scope. One original result of the Open Logic Project unit normal-modal-logic/tableaux, counted once: soundness of tableau derivability for modal entailment. Only the counted corollary and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

Allowed standard prerequisites (editorial).

- Tableau derivability. Gamma derives A when some finite signed set with the negation of A at prefix 1 and the B_i of Gamma at prefix 1 has a closed tableau.. It produces the finite signed set Delta the proof works with (2157-2161).
- Tableau soundness. A signed set with a closed tableau is unsatisfiable; the unit proves this as its soundness theorem.. It contradicts the satisfying interpretation built in the proof (2165-2168).
- Interpretation of prefixes. An interpretation of the prefixes of a branch into a model maps each prefix to a world; the root prefix may be sent to any world.. Mapping prefix 1 to the countermodel world w makes the model satisfy Delta with respect to it (2163-2165).

Source reference. Original theorem thm-tableau-soundness: sources/olp-normal-modal-logic-tableaux/pinned-original-source.tex, source lines 1991–1991. The author proof cites the unit's tableau soundness theorem. The load-bearing work of soundness is carried by that separately proved theorem of the same unit; the counted proof retains its own countermodel construction in full.

Corollary 0.1. *If* $\Gamma \vdash !A$ *then* $\Gamma \models !A$.

Proof. If $\Gamma \vdash !A$ then for some $!B_1$, ..., $!B_n \in \Gamma$, $\Delta = \{\bot(!A)[1], \top(!B_1)[1], \ldots, \top(!B_n)[1]\}$ has a closed !!tableau. We want to show that $\Gamma \models !A$. Suppose not, so for some $M \models$ and w, $M \models !B_i[]w]$ for $i = 1$, ..., n, but $/ \models M[!A][w]$. Let $f(1) = w$; then f is an interpretation of $P(\Delta)$ into $M \models$, and $M \models$ satisfies Δ with respect to f. But by thm-tableau-soundness, Δ is unsatisfiable since it has a closed !!tableau, a contradiction. So we must have $\Gamma \vdash !A$ after all. □

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